JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 85, NO. B7, PAGES 3699–3710, JULY 10, 1980

The Paleomagnetism of Certain Late Precambrian and Early Paleozoic Rocks From the Red Sea Hills, Eastern Desert, Egypt

J. DAVIES

Cities Service Company, Houston, Texas 77002

A. E. M. NAIRN

Department of Geology, University of South Carolina, Columbia, South Carolina 29208

R. RESSETAR

Department of Earth Sciences, University of North Carolina, Wilmington, North Carolina 28403

A stable remanence was isolated from three groups of rocks collected in the Eastern Desert of Egypt. Two groups of dikes, one collected north of the Qena-Safaga road and the other about 200 km to the south near the Um Rus gold mine, gave paleomagnetic pole positions of 87°N, 304°E and 86°N, 185°E, respectively. Both have K/Ar ages in the range 480–530 m.y. The results are compatible with data reported from other Gondwana cratonic areas for this time period. The third group of rocks were samples of the Dokhan Volcanic Series collected at seven sites along the Qena-Safaga road. The paleomagnetic pole position of the higher-coercivity component, 36°S, 17°E, acquired about 600 m.y. ago is not easily reconciled with established polar wander paths. The younger component, however, with a paleomagnetic pole lying at 54°N, 327°E, suggests a Cambro-Ordovician age.

INTRODUCTION

Although the Precambrian rocks of the Egyptian basement crop out in the Nile Valley and at Gebel Oweinat, the frontier region with the Sudan and the Libyan Arab Jamahariya, by far the most extensive outcrops occur in the Red Sea Hills (Figure 1). Here a Precambrian succession extends virtually without break from latitude 30°N to the Sudanese border. To the west the Precambrian rocks plunge beneath the Cretaceous arenaceous beds of the Nubian Formation. The eastern margin of the Red Sea Hills is usually an unconformity between the Precambrian and the Mesozoic-Cenozoic, in some places Campanian-Maestrichtian or Paleocene-Eocene, in others Miocene.

The existence of Precambrian rocks in the Red Sea Hills has been known for a long time, but despite the monumental efforts of *Hume* [1934, 1935] and more recently by Egyptian geologists [e.g., *El Shazly,* 1964], they have remained relatively poorly known, for the numbers working in this extensive but inhospitable terrain has remained small.

The oldest known rocks so far recorded in the Eastern Desert are the migmatized gneisses from the Abu Swayel area which have yielded Rb/Sr ages ranging from 1150 to 1300 m.y. [*El Shazly et al.,* 1973]. The most useful working classification of these rocks, using radiometric ages to establish a sequence of events, is that of *El Shazly* [1977], which modifies an earlier attempt [*El Shazly,* 1964] and which owes a debt to *Akaad and El Ramly* [1960] and to *Sabet* [1972]. *El Shazly* [1977] relates the lithological sequence to a tectonic structural scheme (Table 1). In this scheme the Dokhan Volcanic Formation represents emerging geosynclinal volcanics, and the overlying Hammamat Formation contains primarily arenaceous sediments of a postgeosynclinal phase. However, a Rb/Sr isochron age of 603 m.y. obtained from the Dokhan Volcanics by R. Stern (personal communication, 1979) suggests the lavas are coeval with the lower part of the Hammamat Formation, i.e., a lateral facies equivalent, sedimentation having outlasted volcanism. These formations are cut by late orogenic and postorogenic plutons and subsequently by younger minor intrusions, sills, dikes, and plugs of a variety of ages and compositions. The postorogenic minor intrusions are extremely numerous, and several dike swarms occur, very few of which have been dated.

As a perusal of the literature suggested that none of the sequences younger than the synorogenic plutons were severely metamorphosed, they formed obvious targets for paleomagnetic research. Paleomagnetic investigations of African Precambrian–early Paleozoic rocks have thus far been concentrated in the East African and South African cratons [e.g., *Piper,* 1976]. Establishing magnetic directions for Egyptian rocks would therefore aid in determining the relation of the Nubian craton to those of the rest of Africa.

With these objects in mind, collections in excess of 2000 oriented samples were made in regions accessible from Quseir, the Qena-Safaga, and Idfu-Marsa Alam roads (Figure 1). The first of the results is reported here. Since the interpretation of the paleomagnetic data cannot be separated from the geological controls, it is convenient to preface each of the paleomagnetic sections, which consider separately the Dokhan Volcanics, Hammamat Sediments, and dike swarms, by an outline of the relevant geology.

GEOLOGICAL AND PALEOMAGNETIC OBSERVATIONS

Common to all the groups of rocks sampled were the collection and measurement techniques. For example, most samples were collected as cores drilled in the field; block samples which had to be later cored in the laboratory were only taken when the drills failed. We followed the common technique of collecting approximately six oriented cores per igneous rock unit but only one oriented sample per bed in the sedimentary sequence, the Hammamat Formation. In all cases, rocks were oriented by use of a Brunton compass. Sedimentary samples were collected, maintaining a 1-m sampling interval as far as

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Paper number 9B1687.
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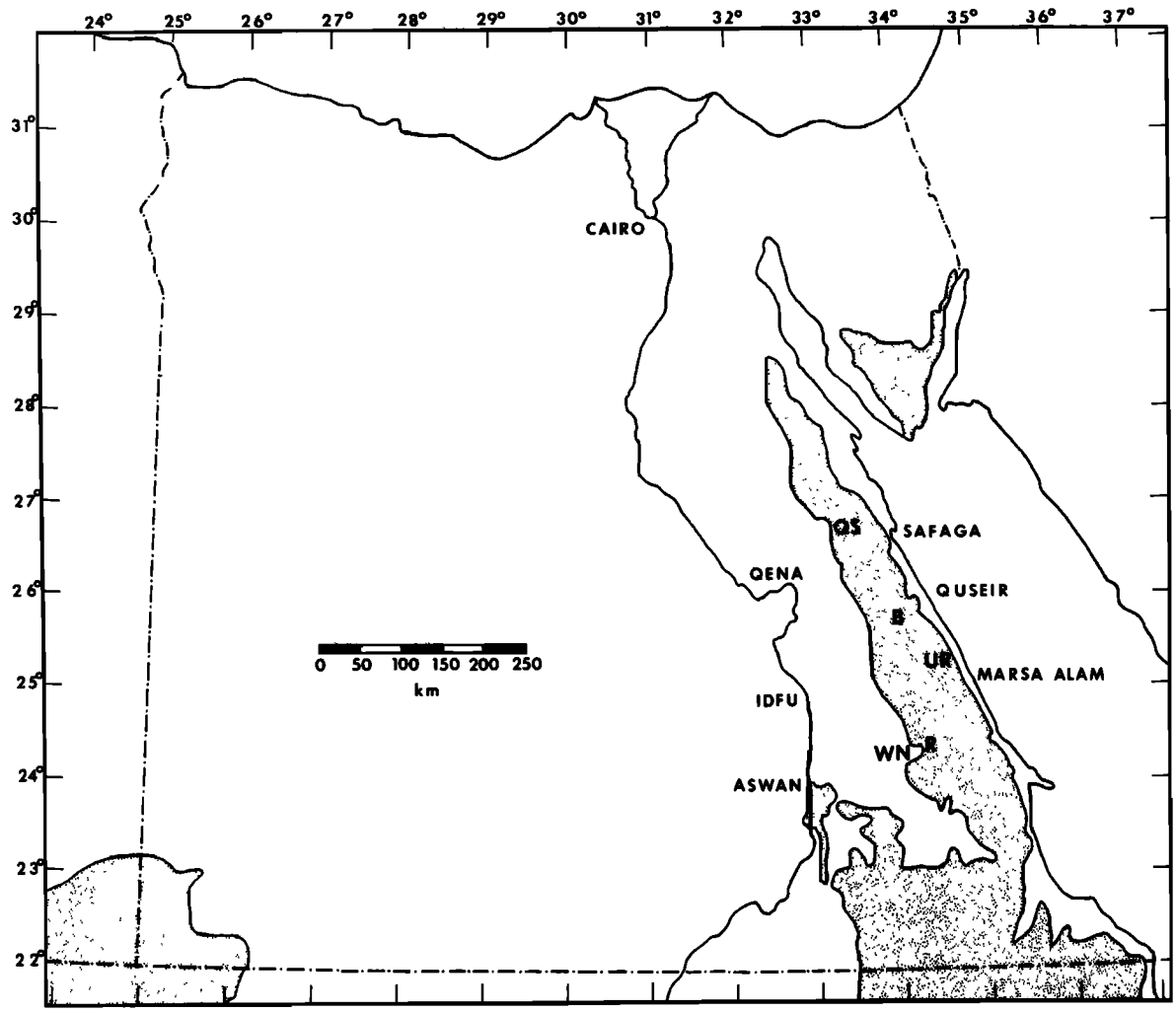


Fig. 1. Map of Egypt showing the outcrop of basement rocks in stippled pattern [after *Akaad and El Ramly*, 1960]. The approximate location of different rock groups is indicated: QS is Qena-Safaga dike swarm, and UR is Um Rus dikes. (B, WN, and R are locations of Cretaceous igneous rocks.)

possible. Directions of and intensity of remanence were calculated from measurements made with a Digico magnetometer; a Schonstedt magnetometer was used only as a means of confirming measurements made on samples with an intensity of magnetization less than 1.0×10^{-6} emu/cm^3 (corresponding to intensities of 0.3–0.4 am^2 kg for rock densities of 2.67–3.1 g/cm^3). Schonstedt thermal and alternating field demagnetizing equipment was available for magnetic cleaning. Before the Schonstedt thermal demagnetizer became available, thermal demagnetization was carried out in a small, noninductively wound furnace housed in a mu-metal room in which the residual field in the furnace was reduced to about 15 γ by the use of ancilliary coils. These progressive demagnetization techniques of magnetic cleaning have been described sufficiently and often, alleviating need for repetition.

The Dokhan Volcanic Formation

General geology. The Dokhan Volcanics form the extrusive element of a phase of igneous activity which *El Shazly* [1977] assigns to the emerging geosynclinal phase. The rocks occur as rhyolites, andesites, and trachytes in the form of flows or sheets in which columnar jointing may sometimes be found. They are associated with volcanic necks, veins, and dikes. They crop out in widely scattered, small outcrops in the Eastern Desert from Gebel Dara (28°N) [*Hume*, 1934] to Wadi Abu Swayel (23°N) [*El Shazly et al.*, 1973]. *Schürmann* [1966] and *Sabet* [1972] have suggested correlations of the Dokhan Volcanics with a wide range of similarly aged rocks and lithologies from Yemen, Saudi Arabia, Sudan, and the Sahara.

The rocks take their name from Gebel Dokhan, the type locality, where the basal part of the volcanics is an ornamental stone. This, the Imperial Porphyry, was well known in classical times and has been described on many occasions [e.g., *Delesse*, 1850; *Rosenbusch*, 1877; *Liebish*, 1877; *Schneider*, 1887; *Barthoux*, 1922; *Hume*, 1934, 1935; *Andrew*, 1938; *Schürmann*, 1966; *Ghobrial and Lofti*, 1967]. In contrast to its general stratigraphic position the absolute age of Dokhan Volcanic Formation is not well known. There are but two Rb/Sr isotopic age determinations which yield dates of 654 m.y. and 665 m.y., respectively [*El Shazly et al.*, 1973], and which were obtained from rhyolites in the Abu Swayel area. The K/Ar date for the underlying Imperial Porphyry (465 m.y.) at Gebel Dokhan given by *El Ramly* [1962] is clearly a minimum age due to argon loss.

It is generally agreed that the rocks have been metamorphosed [*Ghobrial and Lofti*, 1967] and that alteration has not gone beyond a low-grade greenschist phase. *Gindy* [1972] stated that the metamorphism was regional in contrast to *El Shazly* [1977], who regards the alteration as essentially contact

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TABLE 1. Classification of the Basement Rocks of Egypt (Modified From *El Shazly* [1964, 1977])

Structural Stage	Geologic Stage	Type Locality	Radiometric Age, m.y.	Sources of Ages*
Exorogenic foreland (platform)	basic volcanics	Qatrani Volcanics	...	
	alkaline volcanics with subordinate plutonism	Wadi Natash Volcanics	40–80	1, 2
Major unconformity				
Postorogenic preplatform (fourth basement)	postorgenic volcanics	Gebel Um Kibish	290–300	3
	Paleozoic volcanics and dikes	Um Rus Dikes	460–500	4
		Qena-Safaga Dikes	480–530	4
	postorogenic plutons	Aswan Granite	570–590	5, 6, 7
Orogeny (third basement)	late orogenic plutons	younger (pink) granites	480–656	2, 5, 6, 8, 9, 10, 11, 12, 13
Late geosyncline (second basement)	postgeosynclinal sediments	Wadi Hammamat		
	emerging geosynclinal volcanics	Gebel Dokhan Volcanics	603–665	8, 14
	synorogenic plutons	Wadi Garf diorite-granite complex	980	8
Early geosyncline (first basement)	early orogenic plutons	Wadi Shait	876–890	7, 8
	geosynclinal volcanics	Barramiya group (basic), Sheik Shadli group (acidic-intermediate)	856	7
	geosynclinal sediments	Abu Swayel Series	1195	7

*References are 1, *Higazy and El Ramly* [1960]; 2, *Afanassiev* [1960]; 3, *El Shazly* [1964]; 4, *Ressetar* [1979]; 5, *Hashad et al.* [1972]; 6, *Schürmann* [1966]; 7, Pan-American–UAR Oil Company, unpublished data, 1979; 8, *El Shazly et al.* [1973]; 9, *Geith* [1959]; 10, *El Ramly* [1962]; 11, *Schürmann* [1964]; 12, *Fullagar and Greenburg* [1978]; 13, *Hashad* [1978]; and 14, *Stern et al.* [1980].

metamorphism. Whichever may be the case, it is clear that the rocks have been locally crushed, faulted, and tilted [*Schürmann,* 1966]. *Akaad* [1972] regards erosion of the rapidly uplifted Dokhan Volcanic Formation as the source of the material found in the conglomeratic facies of the Hammamat Formation.

Petrology. The rhyolitic, andesitic, and trachytic lavas and tuffs all show some degree of alteration. Some samples contain devitrified glass or a fine-grained groundmass with phenocrysts. The feldspars show the greatest degree of alteration with the development of a chlorite, some quartz, and epidote assemblage. Disseminated opaque minerals are abundant, sometimes exceeding 5%; however, the intensity of magnetization shows that they play little or no part in sample remanence. In reflected light a reddish color is general, often concentrated around grains or in veins. There is a considerable variation from sample to sample, from samples in which red (hematitic?) material obscures the general petrology to those in which alteration is less and original structures and minerals can be recognized.

Paleomagnetic observations. A preliminary suite of samples were collected from seven sites along the well exposed and most accessible section in the Dokhan Volcanic Formation, the Qena-Safaga road, where the rocks dip at an angle of approximately 50° to the south east. This can be seen in certain of the tuff horizons and estimated at other locations by the attitude of the columnar jointng. Detailed consideration of the petrology, geochemistry, and paleomagnetic behavior will appear in a subsequent report based upon current studies of a more complete sampling of the Qena-Safaga section and of exposures in Wadi Queh.

All the sites provided results, but one (site 4, Table 2) was well separated from the remaining six. Of the latter the dispersion of the individual samples resulted in the elimination of site 3, which has an unacceptably high circle of confidence. The natural remanent magnetization (NRM) and cleaned directions of magnetization are listed in Table 2 and are shown in Figure 2.

Thermal demagnetization proved to be the more successful cleaning technique (Figure 3), for the thermal demagnetiza-

TABLE 2. Summary of Paleomagnetic Data From the Dokhan Volcanic Formation Along the Qena-Safaga Road

Site	Rock Type	Number of Samples	Declination, deg	Inclination, deg	α_{95}	K	R	Demagnetization
1	andesite flow	5	190.3	33.4	15.7	24.5	4.84	500°C
2	andesite tuff	5	174.3	47.1	14.7	27.8	4.86	600°C
3	andesite flow	5	165.1	36.0	90	1.3	...	500°C
4	andesite flow	5	39.2	44.8	42.0	2.0	3.02	600°C
5	andesite flow	5	192.9	55.6	9.3	67.9	4.94	600°C
6	andesite flow	6	206.3	40.2	11.5	34.4	5.85	600°C
7	andesite tuff	4	205.6	39.1	10.1	83.6	3.96	500°C
Mean*		5 sites	194.3	43.6	12.1	40.5	4.90	...
8	removed component	6	318.4	54.2	15.3	20.1	5.75	600°C

Paleopole position of hard magnetization, 36.1°S, 17.1°E; *dp,* 14.8; *dm,* 12.1. Paleopole position of soft magnetization (site 8), 53.5°N, 327.0°E; *dp,* 20.5; *dm,* 15.6.

*Sites 3 and 4 not included in the mean.

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tion curve suggests that hematite is the chief carrier of remanence (Figure 4). Only samples from site 6 showed a clear and rapid response to cleaning in alternating fields (Figure 5). At this site the direction of magnetization of a large softer secondary component was isolated (site 8, Table 2). The same direction was observed in samples from the remaining sites, but because of the low intensity of magnetization, results were much more scattered.

Age relations of the Dokhan magnetization. The magnetic cleaning of the Dokhan Volcanic Formation suggests that the remanence is carried by both titanomagnetite and hematite, although the latter seems to be more commonly the case. The presence of both normal and reversed polarities within two of the flows is evidence that the magnetization is secondary. As hematite is more characteristically a secondary mineral in an-

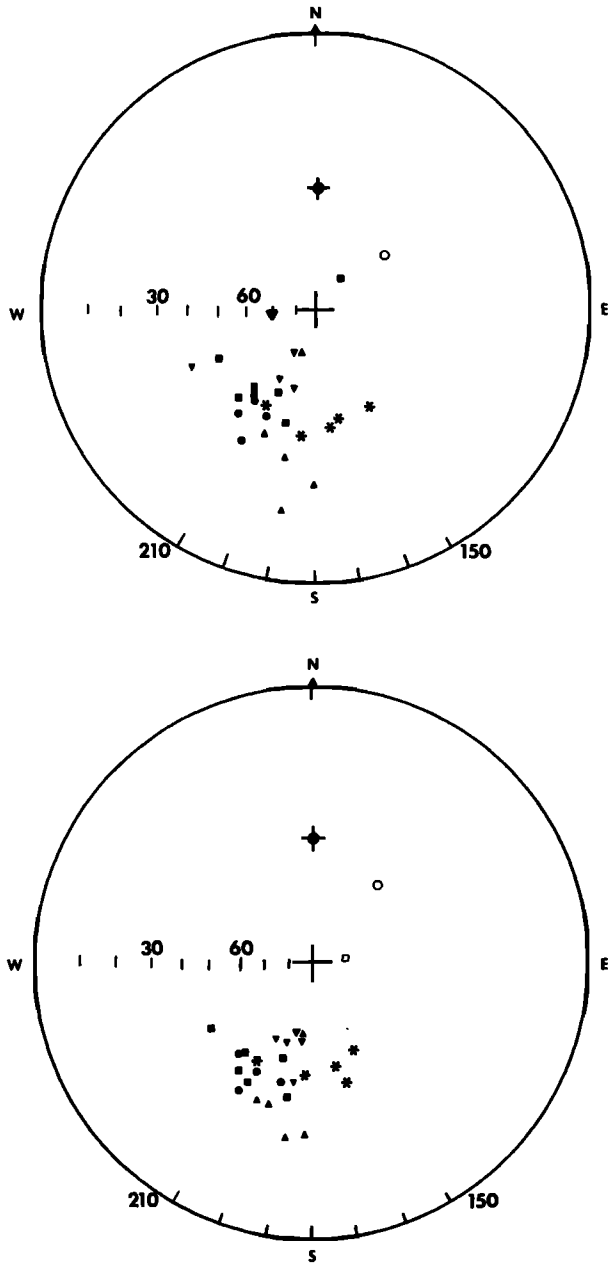


Fig. 2. Stereographic plot of NRM (top) and cleaned (bottom) directions of samples from the Dokhan Volcanics at the Qena-Safaga road section. Open symbols reflect negative inclinations. Site 1, triangles; site 2, asterisks; site 5, inverted triangles; site 6, squares; site 7, circles. Present axial dipole field direction circles with crosses.

desitic rocks, there are two possibilities concerning the age(s) of magnetization, given that the directions of the hard components of both carriers are the same. One is that the hematite magnetization was acquired immediately after emplacement, perhaps from deuteric alteration by circulating hot fluids. Alternatively, if the magnetite is also of secondary origin, both formed at some later, unspecified time.

The thin section study certainly indicates the secondary nature of the hematite. Its association with quartz and epidote suggests that its formation is linked with low-grade metamorphism of the rocks. A similar hematite-quartz-epidote association is also found in the apparently coeval Hammamat sediments (R. Stern et al., personal communication, 1979) but is absent in the late orogenic granites.

The best assessment of the age of the magnetization is thus post-Hammamat but older than the late orogenic plutonites. The reported Rb/Sr ages for the Dokhan Volcanics range from 603 [*Stern et al.*, 1980] to about 660 m.y. [see *Davies*, 1979]. There is a greater spread in the radiometric ages assigned to the late orogenic plutons, for these range from 480 to 621 m.y. More recent work by *Fullagar and Greenburg* [1978] provides whole rock Rb/Sr isochrons in the 580- to 607-m.y. range. The most realistic estimate for the post-Hammamet age of the magnetization is thus about 600 m.y.

There is a minimum K/Ar age, 465 m.y. [*El Ramly*, 1962; *Ghobrial and Lofti*, 1967] for the Imperial Porphyry which forms the lower part of the Dokhan Volcanic Formation. This suggests a possible age for the softer secondary component (site 8, Table 2). As the geomagnetic pole for this component closely approaches others of Cambro-Ordovician age from other parts of Africa, it suggests that there was a minor thermal event at about 500 m.y. following the main event at 600 m.y. This event was largely unrecorded in the rocks where hematite was the carrier of remanence, but in those samples where the carrier was titanomagnetite, its lower coercivity allowed a clear imprint to be imposed. There is some confirmatory evidence for this from Saudi Arabia, where two thermal maxima during the Pan-African event are recognized. The first, more intense, was between 560 and 610 m.y.; the second, with maximum temperatures in the 300°–350° range, occurred between 510 and 540 m.y. [*Fleck et al.*, 1976].

The Hammamat Formation

The postgeosynclinal sediments [*El Shazly*, 1977], grouped into the Hammamat Formation, a name taken from the wadi in which they are particularly well exposed, crop out between Wadi Ranga (24.5°N) and Wadi Dib (28°N). At the type locality neither the base nor top of the formation is exposed [*Andrew*, 1939], but in the Wadi Kareim they can be seen to rest unconformably upon metasediments and metavolcanics of the second basement group [*Abd El Aziz*, 1968], and elsewhere they have been found overlying the Dokhan Volcanic Formation.

The sediments, a molasse sequence of alternating red and green conglomerates, greywackes, mudstones, and siltstones [*El Shazly*, 1977; *Akaad and El Ramly*, 1972], have been separated into a number of members which appear to have little more than local significance [*Grothaus*, 1979]. One of the conglomerate horizons was described as the 'breccia verde antico' by *de Roziere* [1826]. Sediments of apparently similar lithology and age are known in the Sudan, Sinai, Saudi Arabia, Israel, and Jordan [*Schürmann*, 1966; *Karpoff*, 1960; *Brown and Jackson*, 1960; *Geukens*, 1960; *Ruxton*, 1960].

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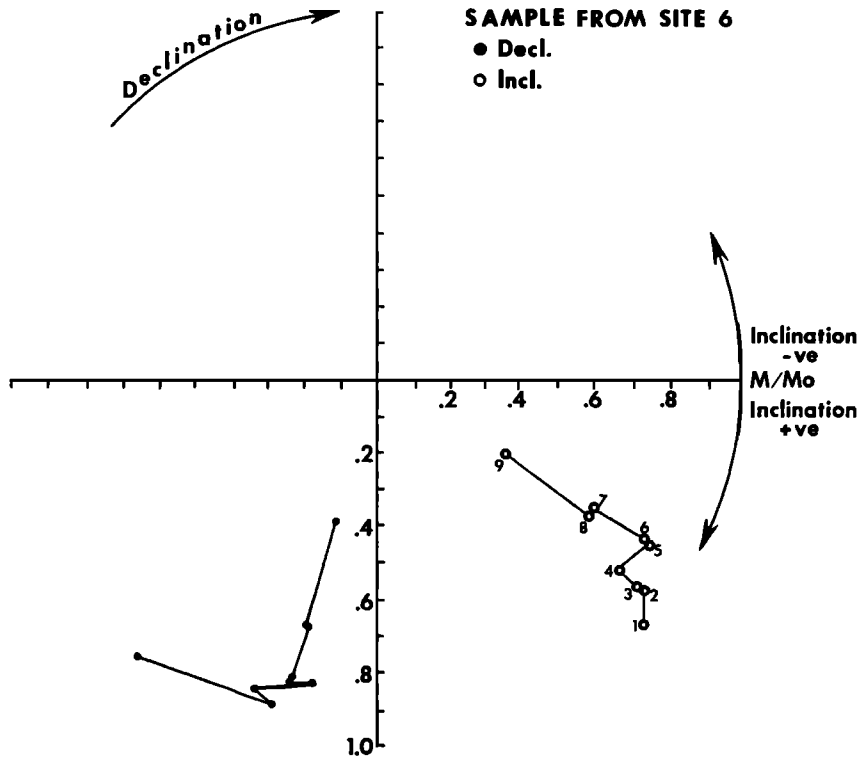


Fig. 3. Stable endpoint diagram in which declination and inclination are plotted against normalized intensity (M/M_0). The sample is from site 6 of the Dokhan Volcanics. Demagnetization stages are 1, NRM; 2, 150 Oe; 3, 200 Oe; 4, 250 Oe; 5, 450°C; 6, 500°C; 7, 550°C; 8, 600°C; 9, 650°C (1 Oe = 0.1 mT).

The Hammamat sediments have been subjected to low-grade regional and contact metamorphism [*Abd El Aziz,* 1968; *Schürmann,* 1966]. This is apparent in thin section, where a hematite-quartz-epidote assemblage similar to that found in the Dokhan Volcanic Formation occurs.

No reliable magnetic data were isolated from the Hammamat Formation. Directions of magnetization remained dispersed both before and after thermal demagnetization and alternating field treatment. No stable endpoint was reached after acid leaching in contrast to the successes shown by *Morris and Carmichael* [1978] and *Roy and Park* [1974].

The Dike Swarms

There are in the Eastern Desert many dike swarms, to the extent that *Vail* [1970] considered it one of the most highly intruded regions of the world. Two of these swarms were sampled. The first consists of a swarm of mafic dikes intruded into a late orogenic granite north of the Qena-Safaga road, and the second, a sequence of dikes of variable composition, was intruded into late orogenic granite and metamorphic rocks in the vicinity of the Um Rus gold mine near the Red Sea coast north of Marsa Alam.

These granites show no evidence of subsequent deformation. Although there are no directly overlying beds, the unconformity at the base of Cretaceous is essentially horizontal in the Red Sea Hills. There is thus no obvious structural control, so that all measurements of remanence were corrected to the present horizontal. The coincidence of the directions of magnetization of the two coeval dike swarms some 200 km apart suggests that any correction would have to be applied to the region as a whole.

The Qena-Safaga (QS) dikes. A general description of and paleomagnetic results from the QS dikes are as follows.

General description: North of the Qena-Safaga road over an area of a few square miles lies a concentrated swarm of NNE trending, easterly dipping mafic dikes. They rise as long

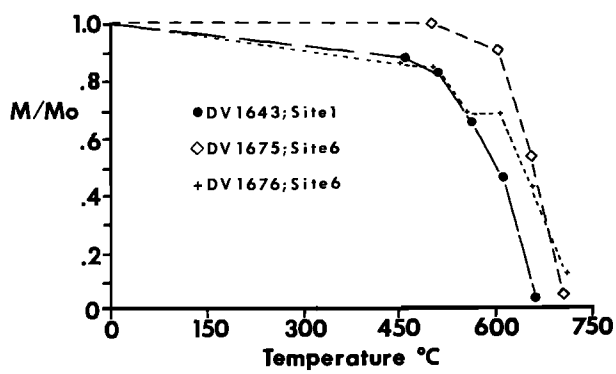


Fig. 4. Thermal demagnetization curves of samples from the Dokhan Volcanics.

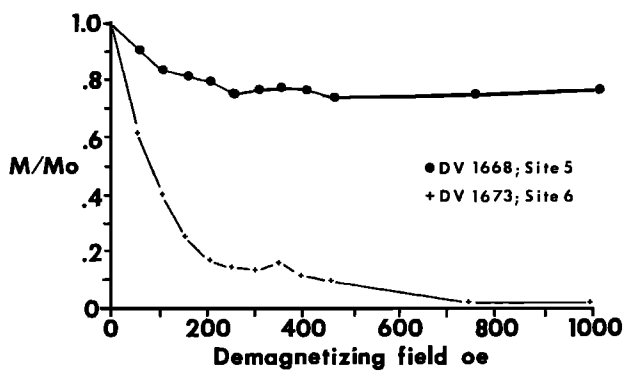


Fig. 5. Alternating magnetic field demagnetization curves of samples from the Dokhan Volcanics (1 Oe = 0.1 mT).

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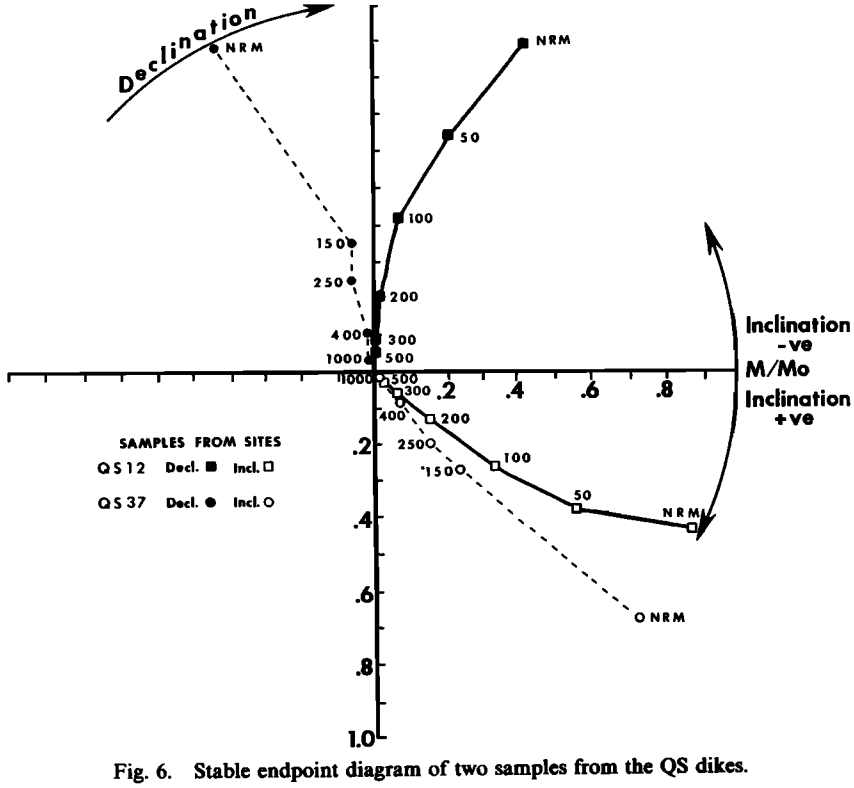


Fig. 6. Stable endpoint diagram of two samples from the QS dikes.

linear mounds from the otherwise featureless sand- and gravel-covered basement. In general, they remain remarkably parallel to one another. The dikes are thin, seldom exceeding 10 m in thickness, and it is quite common to find three or four close together each of the order of 1–2 m in thickness separated by 4–10 m of country rocks.

Microscopic study shows that the dikes are mostly hornblende microdiorites metamorphosed to greenschist facies.

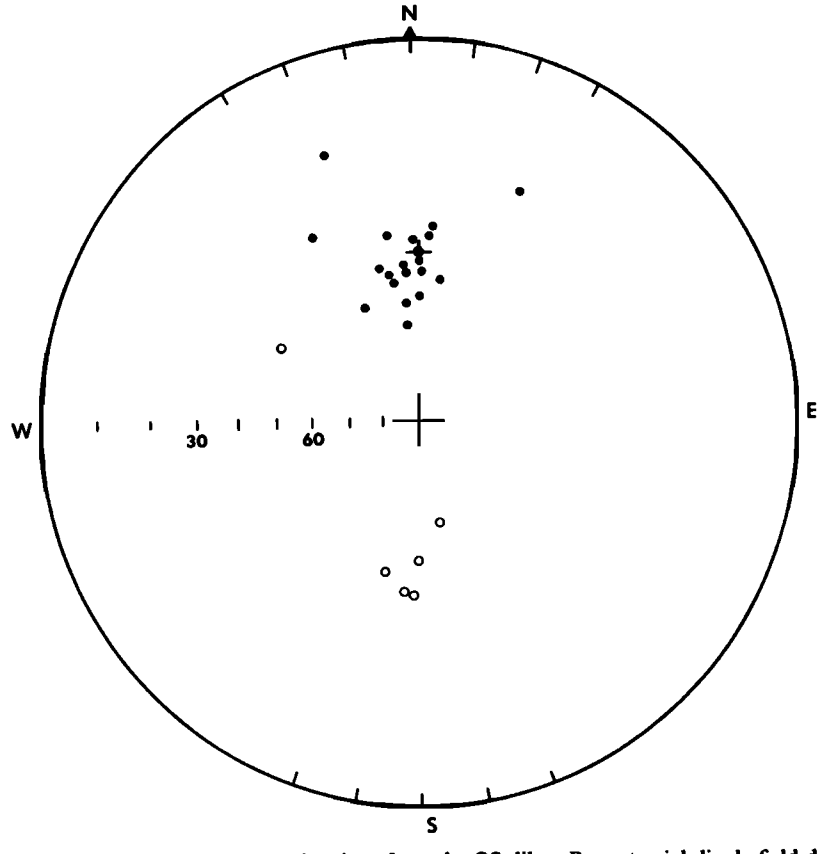


Fig. 7. Stereographic plot of cleaned site mean directions from the QS dikes. Present axial dipole field direction, circle with cross.

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TABLE 3. Summary of Paleomagnetic Data from the QS Dikes

Site	Number of Samples Measured/ Collected	Declination, deg	Inclination, deg	α_{95}	K	R	Demagnetization
QS3	5/5	169.4	−58.9	12.0	41	4.902	400 Oe
QS5	4/4	195.7	−46.3	28.5	11	3.734	550 Oe
QS6	6/6	182.0	−50.3	15.2	20	5.754	250 Oe
QS7	6/6	184.3	−41.4	9.7	49	5.897	300 Oe
QS8	6/6	354.7	+61.7	13.7	25	5.798	200 Oe
QS9	6/6	92.7	+31.7	70.5	2	3.309	NRM
QS11	6/6	179.4	−23.0	42.4	3	4.547	500 Oe
QS12	4/6	348.6	+47.9	15.5	36	3.916	500 Oe
QS13	4/6	349.7	+48.3	14.7	40	3.924	100 Oe
QS15	5/6	326.5	−33.6	40.2	6	5.32	550°C
QS16	6/6	29.8	+26.3	23.9	9	5.32	50 Oe
QS17	5/5	105.8	+2.8	55.4	3	3.602	NRM
QS18	3/5	351.4	+36.5	21.1	35	2.943	200 Oe
QS19	2/3	335.2	+53.0	17.3	27	1.995	200 Oe
QS20	4/6	341.6	+16.6	14.5	41	3.92	100 Oe
QS21	5/6	187.4	−49.5	53.6	3	3.661	450°C
QS22	5/6	1.7	+47.4	7.1	114	4.965	250 Oe
QS25	7/7	323.1	−61.8	33.7	4	5.557	250 Oe
QS28	6/6	215.2	+37.9	69.3	2	3.350	500 Oe
QS30	4/6	3.7	+38.3	14.6	40	3.925	100 Oe
QS31	6/6	359.1	+39.4	4.5	220	5.977	200 Oe
QS33	6/6	271.7	+14.2	33.6	5	4.980	50 Oe
QS34	5/6	0.7	+54.3	8.4	82	4.951	150 Oe
QS35	5/6	187.0	−42.0	23.8	12	4.464	250–350 Oe
QS36	6/6	4.8	+35.6	7.7	76	5.934	150 Oe
QS37	6/6	1.1	+42.1	5.0	174	5.971	150 Oe
QS38	5/6	9.6	+48.7	18.3	18	4.781	250 Oe
QS39	4/5	298.5	−54.7	22.1	18	3.835	350°C
QS41	6/6	1.3	+44.3	5.5	149	5.966	100 Oe
QS42	5/5	49.2	+45.2	49.2	3	3.812	100 Oe
QS43	4/4	355.3	+45.3	10.3	80	3.962	150 Oe
QS44	5/6	355.1	+45.1	9.5	66	4.939	250 Oe
QS45	4/6	321.9	+24.2	30.0	10	3.709	400°C
QS46	6/7	354.2	+55.2	13.1	27	5.814	150 Oe
QS47	5/6	331.2	+31.8	22.9	12	4.669	300 Oe
QS48	5/6	134.9	+29.2	53.2	3	3.674	250 Oe
QS49	5/5	11.5	+62.7	39.9	5	4.133	500 Oe
QS50	6/6	346.5	+44.7	7.4	82	5.939	500 Oe
Mean*	25 sites	357.5	+44.7	4.8	36	24.334	...

Paleopole Position: 87.3°N, 304.2°E; *dp*, 5.0; *dm*, 4.8.
*Site 39 and those with $\alpha_{95} \geq 30°$ not included in the mean.

TABLE 4. Summary of Paleomagnetic Data From the UR Dikes

Site	Number of Samples Measured/ Collected	Declination, deg	Inclination, deg	α_{95}	K	R	Demagnetization, Oe
1	7/8	357.8	43.1	4.7	165.2	6.9637	150
2	4/5	2.3	34.8	25.0	14.4	3.7926	100
3	6/6	7.2	26.3	18.3	14.2	5.6491	150
4	4/4	354.1	32.6	19.2	23.7	3.8736	200
5	8/8	199.1	−35.3	7.5	54.6	7.8718	200
6	5/6	360.0	37.3	5.0	233.5	4.9829	200
11	14 separate units	0.3	36.5	4.1	123.0	10.919	150–250
12	6/6	351.7	39.2	6.5	106.0	5.9530	150
16	6/6	7.3	44.9	6.9	93.7	5.9467	200
17	6/6	4.1	61.9	7.1	87.6	5.9430	200
18	5/5	3.7	49.7	7.4	106.0	4.9623	100
19	6/6	2.5	38.3	5.2	165.0	5.9697	200
20	5/5	21.7	23.2	5.9	166.0	4.9759	300
21	6/6	0.3	33.3	5.7	134.4	5.9628	300
All sites	14 sites	4.0	38.5	5.7	48.3	13.7309	...

Paleopole position: 85.9°N, 193.0°E; *dp*, 6.2; *dm*, 5.4.

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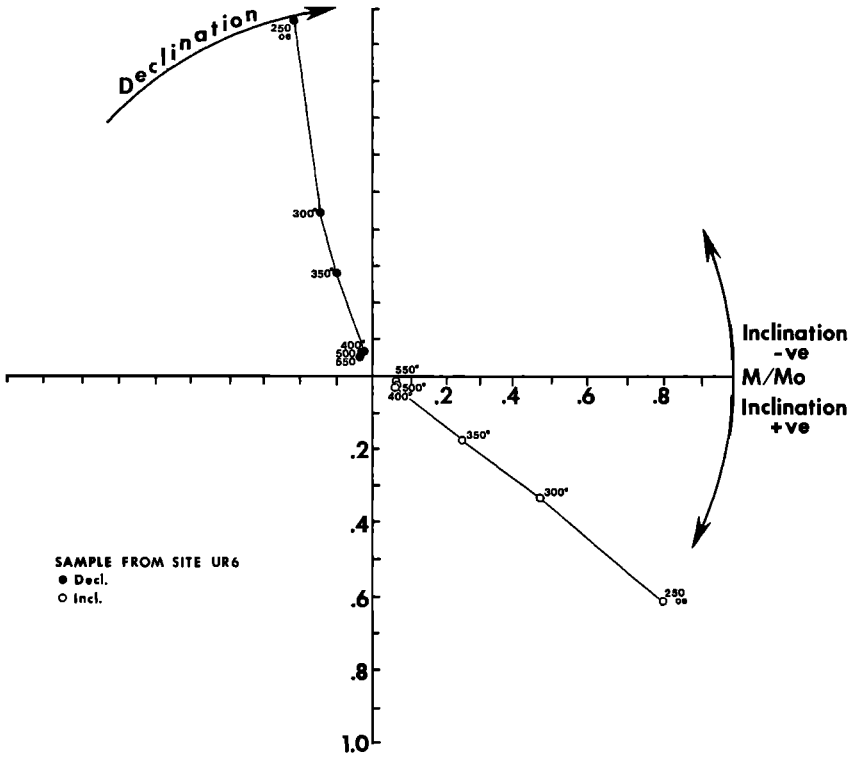


Fig. 8. Stable endpoint diagram of a sample from site 6 of the UR dikes. The intensities at the thermal demagnetization steps are normalized to that measured after af demagnetization at 250 Oe (25.0 mT).

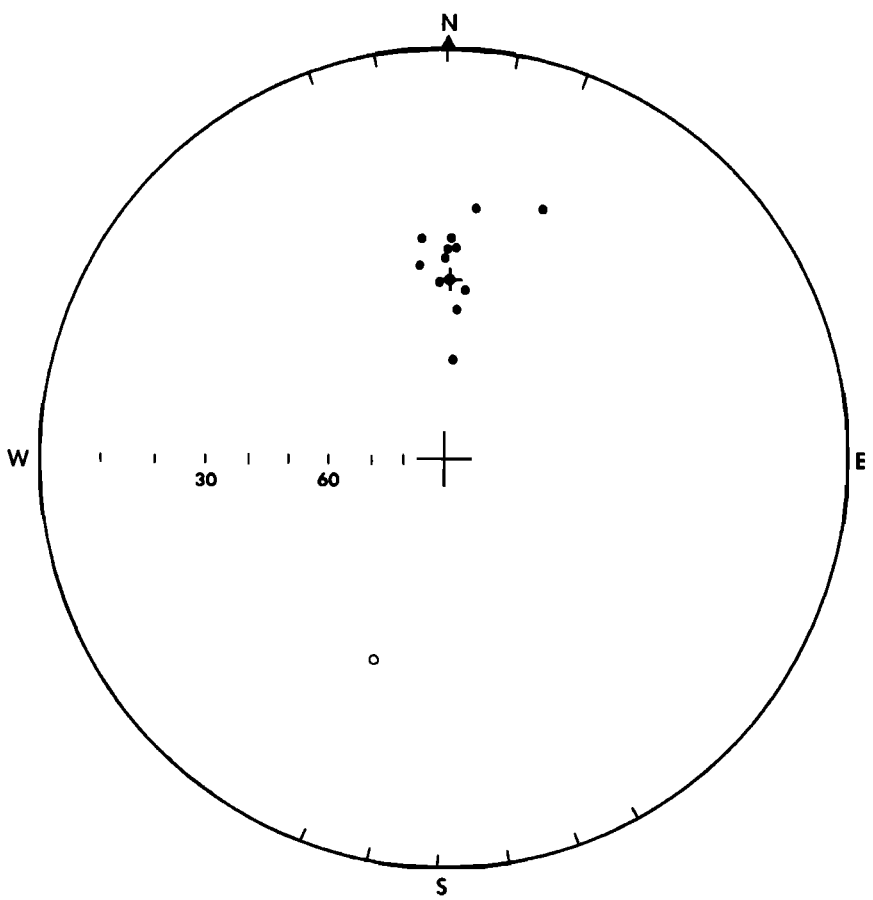


Fig. 9. Stereographic plot of cleaned site mean directions from the UR dikes. Present axial dipole field direction, circle with cross.

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TABLE 5. K-Ar Age Data From QS and UR Dikes

Sample	Site	Weight Percent K	Volume $^{40}Ar_{rad}$, nl/g	$\frac{100 \times {}^{40}Ar_{rad}}{{}^{40}Ar_{total}}$	Age, m.y.
EG-442A	QS3	1.30	30.9465	88.87	527
EG-616	QS32	1.79	42.1391	91.89	521
EG-379	UR19	0.38	8.5363	86.53	497
EG-687	QS44	1.14	24.7373	83.96	488
EG-503A	QS13	1.64	35.7423	88.63	487
EG-641	QS36	1.10	23.4357	84.83	480
EG-316	UR11	1.45	30.4249	82.89	473
EG-344	UR13	1.28	26.9056	71.68	464
EG-358	UR15	3.23	57.7460	84.68	410

The most abundant mineral is plagioclase, which is typically zoned and sericitized. Composition estimated by the Michel-Levy method is that of an andesine or low-calcic labradorite. Small euhedral cyrstals of hornblende are common as well as their alteration products, epidote, actinolite, and sphene. Interstitial quartz grains occur, and apatite and calcite are present in minor amounts. According to I. El Kassas (personal communication, 1976), olivine phenocrysts with reaction rims are found, although none were observed in this investigation.

Paleomagnetic results: Samples were collected from 38 sites. This represented a collection of more than 38 dikes, since, where multiple dikes were present, each was sampled but kept within the single-site nomenclature. Intensities of natural remanent magnetization were often high, ranging up to 5×10^{-2} emu/cm^3. Invariably, these intensely magnetized samples were either magnetically unstable or yielded anomalous directions of magnetization. They seldom responded to magnetic cleaning. Such high intensities of magnetization may be attributed with some degree of confidence to lightning remagnetization.

Most samples responded to cleaning in alternating fields with intensities typically dropping to less than one third after treatment in fields of 200 Oe (20 mT) (Figure 6). The response to alternating field demagnetization was variable, from directions which remained virtually unchanged up to the highest fields possible with the apparatus (1000 Oe) to others which changed continuously without ever reaching a stable endpoint. Many, however, responded well, and the results of these measurements are depicted in Figure 7 and listed in Table 3. An estimate of the number of samples which did not respond to magnetic cleaning can be gauged from the number of samples reported compared to the number available as listed in Table 3, column 2. The stage of demagnetization considered as best representing the original magnetization was selected by choosing the least dispersion value of the precision parameter.

The attempt was made by thermal demagnetization to isolate a stable remanence in those cases where alternating field methods had failed. Although in a few cases a slight decrease in the directional scatter was achieved, in no case did it prove possible to determine the stable component.

The Um Rus (UR) dikes. A general description of and paleomagnetic results from the UR dikes are as follows.

General description: The general geology of the Um Rus area has been drawn from the description of *Kabesh et al.* [1967]. The Um Rus gabbro and late orogenic Gebel Kadabora granite were intruded into the metamorphic basement of Um Rus. A granodioritic-tonalite plug was subsequently injected into the gabbro massif, the contact between the two being marked by a zone of hybrid rocks. All of these units were then cut by a variety of steeply dipping felsic, intermediate, and mafic dikes trending north to northeast. The gabbro is composed primarily of plagioclase, with strongly pleochroic red-brown hornblende and subcalcic clinopyroxene. Chlorite and ferroactinolite are common alteration products. Some of the dikes appear to be offshoots of the gabbro. There are two varieties of felsic dikes, one reddish porphyritic rhyodacite and the other light grey in color, which seems of a more intermediate chemical character.

In general, this southern group of dikes are more altered than those along the Qena-Safaga road. *Kabesh et al.* [1967] have reported quartz-epidote veins which they ascribed to retrogressive hydrothermal alteration.

Paleomagnetic results: Paleomagnetic samples were obtained from the gabbro (site 5), from dikes presumed to be from the gabbro (sites 1–4), from felsic and mafic dikes from near the hybrid zone, from an intricate network of anastomosing basic dikes penetrating the granite (site 11), from a series of parallel, regularly spaced dikes near the Um Rus mine (sites 12–20), and from dikes further south along the Marsa Alam-Idfu road (sites 21 and 22).

The response of the mafic rocks to thermal and alternating field treatment was much the same as the QS series. A range of blocking temperatures was apparent. The results are summarized in Table 4, and the stable end point curve (Figure 8) gives an example of their behavior during cleaning. Figure 9 is a stereogram of the cleaned results. The felsic dikes in general gave less consistent results with higher dispersion.

The age relationship of the dike swarms. The ages of the dike swarms in the Eastern Desert are generally poorly known. Some were considered as late Precambrian and others as Lower Paleozoic. As a result, K/Ar whole rock age deter-

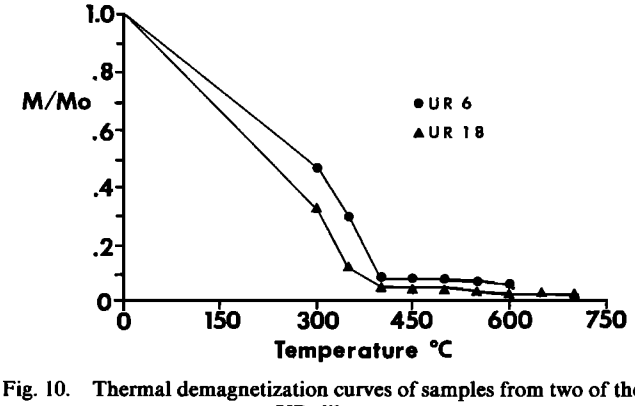


Fig. 10. Thermal demagnetization curves of samples from two of the UR dikes.

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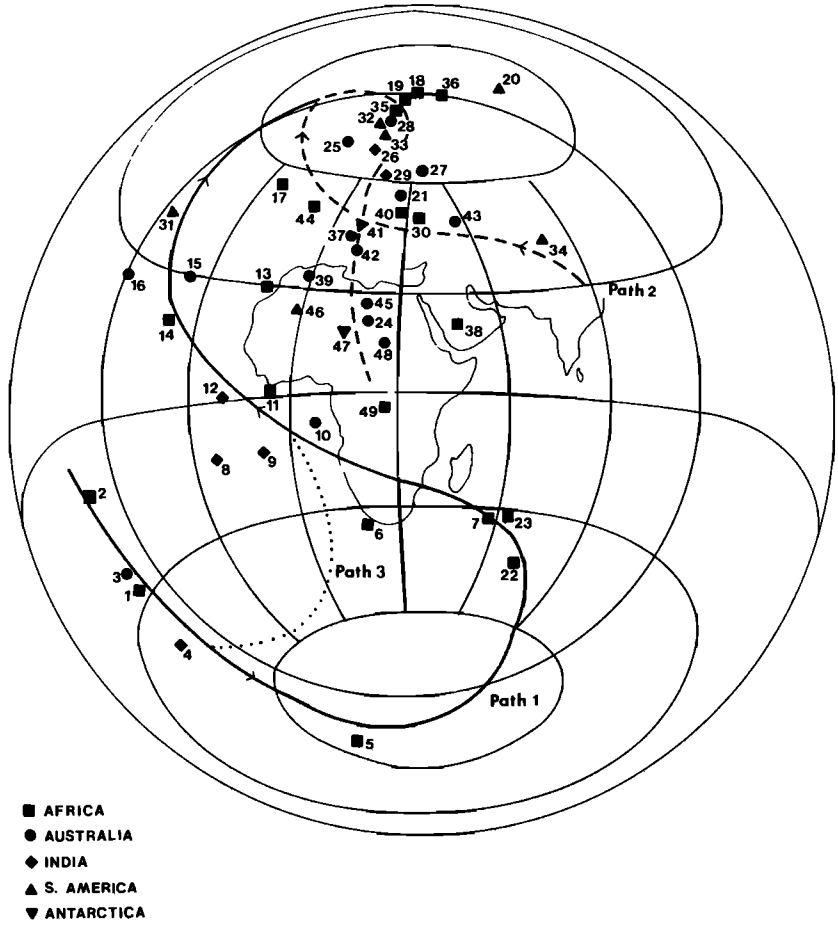


Fig. 11. General trend of the African apparent polar wander path for the late Precambrian and early Paleozoic. The data for the pole positions as numbered are given in Table 6.

minations were undertaken with the results shown in Table 5. Since low-grade hydrothermal alteration has occurred, the ages are best regarded as minimum ages, more accurately reflecting the age of metamorphism than the time of dike emplacement.

We conclude that the dike intrusions occurred shortly after the emplacement of the late orogenic granites and that they are thus related to the last stages of the Pan African event. The effects of a thermal event obviously leave open the question of the age of magnetization.

The thermal demagnetization studies show thermally distributed blocking temperatures (Figure 10). Although in most instances the carrier of remanence appears to be attributable to a titanomagnetite, the persistence of remanence at 600°C suggests that in some cases at least, hematite is important. The geomagnetic pole positions from the two dike swarms lie in high latitudes and could therefore represent either a Cenozoic or an early Paleozoic age. We favor the latter in the absence of any Cenozoic effect being observable in the Cretaceous intrusions found in close proximity to some of the intrusions. This interpretation is consistent with that drawn from the samples of the Dokhan Volcanic Formation mentioned earlier.

DISCUSSION

Of the several attempts to summarize the paleomagnetic pole positions of African rocks of late Precambrian and early Phanerozoic age in terms of 'polar wander' paths, two of the more recent are shown in Figure 11 as paths 1 and 2. These curves, path 1 [*McElhinny and Embleton,* 1976] and path 2 [*Morel and Irving,* 1978], use data from all Gondwana continents (Table 6), thereby implying a unity of Gondwana at this time, a question discussed by *Piper* [1976] and *Nairn* [1975].

The data presented in this paper are shown in Figure 11 as poles 6, 17, 35, and 36. The paleomagnetic poles for the two dike swarms (35 and 36) and the younger component from the Dokhan Volcanics (17) may be fitted to either polar wander path. The older component of the Dokhan Volcanics (6) is more readily interpreted in terms of the McElhinny and Embleton path. This path, however, implies extremely rapid polar wander during the late Precambrian–earliest Paleozoic. This difficulty may be circumvented to some degree by drawing a shorter path (path 3) through the South Atlantic, following *Hailwood*'s [1972] suggestion that poles 7, 22, and 23 represent a Hercynian magnetic overprint. It underlines the need for more data to add precision to this interesting segment of the polar wandering path.

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TABLE 6. Late Precambrian to Mid-Paleozoic Paleomagnetic Data from Gondwanaland

Pole	Rock Unit and Location	Age, m.y.	Pole Position, deg		α_{95}	Reliability	Reference
			Latitude	Longitude			
1	Gagwe Lavas, Tanzania	813± 30	29°S	283°	11	A	P-45
2	Bukoban Intrusions, Tanzania	806 ± 30	11°S	281°	16	A	P-44
3	YB Dikes, Australia	750 ± 30	26°S	296°	...	B	M-YB
4	Malani Rhyolites, India	745±10	40°S	282°	...	A	M-MR
5	Mbozi Complex, Tanzania	743 ± 30	72°S	248°	14	A	P-47
6	Dokhan Volcanics,‡ Egypt	~603	36°S	17°	8	A	4
7	Ourzazate Volcanics Morocco*	Pr/C	30°S	57°	17	A	1
8	Bhander Sandstone, India	Pr/C	15°S	335°	...	...	M-BH
9	Upper Rewa Sandstone, India	Pr/C	11°S	350°	...	...	M-UR
10	Pound Quartzite, Australia	latest Pr	5°S	6°	24	B	M-PQ
11	Sijarira Group, Rhodesia	Pr/C	2°N	352°	18	A	P-51
12	Upper Bhander Sandstone, India	Pr/C	4°N	339°	...	...	M-UB
13	Ntonya Rings, Malawi	630 ± 24	28°N	345°	2	A	P-50
14	Klipheuval Formation, South Africa	Pr/C	16°N	316°	...	...	P-52
15	Antrim Plateau Volcanics Australia	Pr/C	26°N	319°	13	A	M-Av
16	Arumbera Sandstone, Australia	Pr/C	24°N	294°	16	A	M-AS
17	Dokhan Volcanics,‡ Egypt	~500	54°N	327°	15	B	4
18	Pre-Nama Dikes,† South Africa	653 ± 70	85°N	228°	25	B	P-49
19	Lower Buanji Series, Tanzania	Pr/C	87°N	263°	7	A	P-48
20	Plateau Series A, Zambia	Pr/C	71°N	173°	15	A	M-P7A
21	Plateau Series B, Zambia	Pr/C	60°N	25°	13	A	M-P7B
22	Amouslek Tuffs,* Morocco	Cl	42°S	70°	13	A	1
23	Serie Lie de Vin,* Morocco	Cl	29°S	63°	21	B	2
24	Aroona Dam Sediments, Australia	Cl	21°N	20°	17	A	
25	Hudson Formation, Australia	Cl	72°N	338°	13	A	M-HF
26	Purple Sandstone, India	Cl	71°N	357°	...	...	M-PS
27	Hugh River Shale, Australia	Cl–m	68°N	33°	9	B	M-HS
28	Sabaloka Ring, Sudan	540	83°N	339°	...	...	P-54
29	Salt Pseudomorph Beds, India	Cm	66°N	13°	...	...	M-SP
30	Moroccan Lavas, Morocco	Cm	53°N	34°	...	...	M-MRL
31	Purmamarca Village, Argentina	C	36°N	295°	26	B	3
32	South Tilcara Sediments, Argentina	C	82°N	329°	23	B	3
33	North Tilcara Sediments, Argentina	C	80°N	350°	23	B	3
34	Purmamarca Sediments, Argentina	C	45°N	80°	14	A	3
35	QS Dikes, Egypt	480–527	87°N	304°	5	A	4
36	UR Dikes, Egypt	464–497	86°N	185°	6	A	4
37	Lake Frome Group, Australia	Cm–Ol	43°N	8°	13	A	M-LFG
38	Doornport Formation, South-West Africa	500–550	22°N	45°	9	A	P-57
39	Dundas Group, Australia	Cu	31°N	358°	...	...	M-DG
40	Tassili Sediments, Algeria	C–O	53°N	26°	6	...	P-60
41	Charnockites, Antarctica	C–O	49°N	14°	...	...	M-C
42	Jinduckin Formation, Australia	Ol	44°N	10°	11	A	M-JF
43	Stairway Sandstone, Australia	Om	51°N	49°	9	...	M-SS
44	Table Mountain Series, South Africa	O	50°N	349°	...	...	P-59
45	Tumblagooda Sandstone, Australia	O	27°N	18°	9	A	M-TS
46	Salta and Jujuy Redbeds, Argentina	O	22°N	358°	16	A	M-SJ
47	Sor Rondane Intrusion, Antarctica	485 ± 25	18°N	15°	6	A	M-SRI
48	Mereenie Sandstone, Australia	S?–D	16°N	25°	11	...	M-MS
49	Msissi Norite, Morocco	D	1°S	25°	17	A	1

Ages are Pr, Precambrian; C, Cambrian; Cl, Lower Cambrian; Cl–m; Lower to Middle Cambrian; Cm, Middle Cambrian; Ol, Lower Ordovician; Cu, Upper Cambrian; O, Ordovician; Om, Middle Ordovician; S, Silurian; D, Devonian. Pole positions are listed after rotation to account for post-Paleozoic drift as given by *McElhinny and Embleton* [1976]. All longitudes are east. Reliability criteria are those of *Hicken et al.* [1972]. References: P-00 refers to the pole list of *Piper* [1976]; M-XX refers to the pole list of *McElhinny and Embleton* [1976]; 1, *Hailwood* [1972]; 2, *Martin et al.* [1978]; 3, *Thompson* [1973]; 4, this paper.

*Possibly Hercynian remagnetization.

†K-Ar age of one site is 401 m.y.; others may be of Karoo age.

‡The age of the Dokhan Volcanic Formation and of the magnetization of this formation are estimated on a combination of geological and paleomagnetic criteria.

Acknowledgments. This research was made possible through NSF grant INT 78-01469 to W. H. Kanes, University of South Carolina. We owe a deep debt of gratitude to our many Egyptian colleagues who facilitated the work and would like to mention particularly E. M. El Shazly for his inspiration and I. El Kassas for his tireless enthusiasm and field guidance. We are also grateful to J. M. Wampler, Georgia Institute of Technology, for his help and the use of his laboratories for K/Ar dating of samples. Help with measurements given by M. Ward, M. A. Langford, A. Maynard, W. van Buren, H. Kemmerlin, and I. Talleyvast is gratefully acknowledged.

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(Received August 10, 1979;
revised October 16, 1979;
accepted November 6, 1979.)

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